

The second moment operator determines quantum kernel trainability, expressivity, and sample efficiency

[Authors]

Summary. Quantum kernel methods promise exponential feature spaces but suffer from barren plateaus—the exponential vanishing of gradients that makes training impossible at scale. Here we prove that the second moment operator M_2 , a positive operator on the doubled Hilbert space whose spectrum encodes concentration, trainability and simulability simultaneously, determines which quantum circuits can be trained and which cannot. We establish a gradient variance identity $\text{Var}[\partial L_{\text{reg}}/\partial\theta] \geq (\sqrt{\varepsilon} - \lambda\sigma)^2$ showing that local purity regularization provides a universal, non-vanishing gradient bound, and demonstrate that M_2 ’s spectral rank predicts classification accuracy without running the classifier (216 circuit configurations evaluated in 75 seconds). Empirically, M_2 -selected circuits achieve 99.3% accuracy from just 50 training samples where classical methods require 500, and maintain stable performance across qubit counts $n = 4\text{--}20$ with local-to-global gradient variance ratios exceeding 10^9 . However, this advantage is confined to discrete classification on geometric features: regression, temporal prediction and contact-rich tasks all fail pre-verification. We provide a precise mathematical characterization—grounded in the fibre bundle geometry of quantum circuits—of where quantum kernels help and where they do not. The entire formalization comprises 572 Lean 4 declarations with zero `sorry`, verified across six independent tools.

Quantum machine learning seeks to exploit the exponential dimension of quantum Hilbert spaces for computational advantage^{1–3}. Parameterized quantum circuits encode classical data x into quantum states $\phi(x) = U(x)|0\rangle^{\otimes n}$, inducing a kernel $K(x, y) = |\langle\phi(x)|\phi(y)\rangle|^2$ that operates in a 2^n -dimensional feature space^{4,5}. In principle, this exponential richness should enable learning tasks intractable for classical methods. In practice, two pathologies frustrate this promise: *barren plateaus*^{6,7}, where gradients vanish as $O(1/4^n)$, making training impossible; and *classical simulability*⁸, where circuits that avoid barren plateaus can be efficiently simulated classically, eliminating any quantum advantage. This tension—trainability versus advantage—is the central dilemma of quantum machine learning^{9,10}.

We resolve one side of this dilemma by identifying the single mathematical object that governs both phenomena: the *second moment operator* $M_2 = \mathbb{E}_\theta[U(\theta)^{\otimes 2} \otimes U(\theta)^{* \otimes 2}]$, a positive operator on the doubled Hilbert space $\mathcal{H} \otimes \mathcal{H}$. For Haar-random circuits, M_2 decomposes on the Schur–Weyl subspaces $\text{Sym}^2(\mathcal{H}) \oplus \Lambda^2(\mathcal{H})$ with eigenvalues determined by the Weingarten function¹¹: $\lambda_{\text{sym}} = 1/(d(d+1))$, $\lambda_{\text{anti}} = 1/(d(d-1))$ where $d = 2^n$. The spectral gap of M_2 controls concentration⁶; its rank equals the dimension of the dynamical Lie algebra⁹; and its restriction to local operators determines trainability⁷. We prove these connections formally and show that M_2 serves as a practical circuit design tool: computing its rank for 216 circuit configurations (8 single-qubit gate sets $\times$ 9 two-qubit gates $\times$ 3 depths) takes 75 seconds and predicts classification accuracy without training any model (Fig. 1).

The gradient variance identity. The gradient of a regularized loss $L_{\text{reg}} = L_{\text{task}} + \lambda R$ satisfies

$$\text{Var}\left[\frac{\partial L_{\text{reg}}}{\partial \theta}\right] \geq \varepsilon + \lambda^2 \sigma^2 - 2\lambda\sigma\sqrt{\varepsilon} = (\sqrt{\varepsilon} - \lambda\sigma)^2, \quad (1)$$

where $\varepsilon = \text{Var}[\partial L_{\text{task}}/\partial \theta]$ and $\sigma^2 = \text{Var}[\partial R/\partial \theta]$. This identity factors as a perfect square, making non-negativity manifest. When R is the local purity regularizer $R = \sum_k \text{Tr}[\rho_k^2]$ (a sum of single-qubit purities), we establish that $\sigma^2 \geq c > 0$ for a universal constant c , independent of n (Extended Data Table 1). The bound follows from the light-cone structure of local observables⁷: only $O(\text{poly}(n))$ gates affect each single-qubit purity, preventing exponential concentration. Numerically, $\sigma^2 \approx 0.016$ across $n = 4$ –20 (Fig. 2), while the global fidelity gradient decays as $O(1/4^n)$ with measured slope -1.47 (theory: $-\ln 4 = -1.386$). At $n = 20$, the local-to-global ratio exceeds 10^9 .

Circuit selection via M_2 . The commonly used Ry+CZ ansatz generates only the orthogonal group $O(d)$, not the full unitary group $U(d)$, because both gates have real matrix entries. Its M_2 rank is 124/256—it accesses only half of the doubled Hilbert space. We discovered this through M_2 ’s spectral analysis and confirmed it by computing M_2 explicitly for $n = 4$ qubits via Monte Carlo (2,000 samples per circuit). Circuits requiring two rotation axes (Rx+Ry, Ry+Rz, or the full U3 decomposition) achieve rank 256/256 and Haar distance < 0.07 , compared to 124/256 and distance > 0.10 for Ry+CZ. This rank difference translates directly to classification accuracy: full-rank circuits achieve 88–91% on standard benchmarks at $n = 4$, while half-rank circuits achieve 78% (Fig. 1).

Sample efficiency. M_2 -selected circuits exhibit striking data efficiency on classification tasks. On the two-moons benchmark at $n = 8$, quantum kernels achieve 99.3% accuracy from $N = 50$ training samples, outperforming RBF-SVM (90.3%) and XGBoost (86.0%) by 9–13 percentage points (Fig. 3). As training data grows, classical methods improve to 99.7% at $N = 2,000$ while quantum saturates at 99.3% (limited by the $O(N^2)$ kernel matrix). The crossover occurs at $N \approx 500$. This advantage holds across five datasets (two-curves, circles, hidden manifold, iris, digits) and is consistent across three M_2 -selected circuits (Extended Data Table 2).

The sample efficiency advantage extends to reinforcement learning. On tic-tac-toe, a quantum kernel value function reaches 80.5% win rate from 10 training games and *remains constant* regardless of additional data, while classical RBF oscillates between 59–80% and MLP never reaches 70% (Extended Data Table 3). On Connect Four, quantum leads at $N = 10$ (72% vs 65%) but classical overtakes at $N > 50$. The quantum kernel’s Hilbert space geometry provides an implicit regularizing prior: in a fixed reproducing kernel Hilbert space, the optimal predictor lies in the span of training kernels (the Representer Theorem), and adding data can only refine the coefficients, not destabilize them.

The boundary: where quantum kernels fail. Equally important is a precise characterization of where quantum kernels do *not* help. We conducted systematic pre-verification experiments¹² before every planned experiment, computing kernel–task-similarity correlations and proceeding only when $r > 0.5$. Three categories of tasks consistently failed:

Continuous regression. Kernel ridge regression for predicting smooth value functions shows quantum $R^2 = 0.29$ versus RBF $R^2 = 0.67$ at $N = 10$ (quantum is $2.2\times$ worse). The gap persists at all data sizes. The fidelity kernel $|\langle \phi(x)|\phi(y) \rangle|^2$ creates piecewise-constant interpolants on the Voronoi cells of training points, whereas the Gaussian kernel creates smooth interpolants that match smooth target functions.

Temporal/dynamic predictions. Encoding robot trajectories as multi-layer quantum circuits (each timestep $\rightarrow$ one circuit layer) produces kernel–outcome correlation $r = 0.019$ —near zero. All classical methods also fail ($r < 0.04$), indicating the features, not the kernel, are the bottleneck.

Contact-rich tasks. Pushing similarity in a robotic manipulation setting gives $r = 0.21$ for quantum, $r = 0.36$ for RBF—both below the viability threshold. Push outcomes depend on contact dynamics that neither kernel captures from kinematic features.

The pattern is precise: quantum kernels excel at **static, discrete, geometric** classification in **low-data** regimes ($N \leq 100$). They fail at continuous regression, dynamic temporal prediction, and tasks requiring physics beyond kinematics. This boundary is not empirical accident but mathematical consequence: the fidelity kernel’s discrete Hilbert space geometry creates sharp decision boundaries (ideal for classification) that are too sharp for smooth function approximation.

Geometric interpretation. The Q-JEPA (Quantum Joint Embedding Predictive Architecture) framework provides a geometric interpretation of these results. The quantum encoder, predictor and kernel form a fibre bundle $(X, \mathcal{H}, U(d))$ where the base space is the input data, the fibre is the Hilbert space, and the connection (predictor) parallel-transport quantum states. The Q-JEPA energy $E(x, y) = \|\phi(y) - V(\phi(x))\|^2$ is the holonomy defect—the curvature of this bundle. Collapse (all encodings equal) corresponds to zero curvature; the purity regularizer penalizes flat connections, preserving non-trivial geometry. The dynamical Lie algebra dimension $\dim(\mathfrak{g})$ serves as an order parameter: polynomial dimension yields trainability, exponential dimension yields barren plateaus, and G -equivariance restricts $\dim(\mathfrak{g})$ by a factor of $|G|$. These connections are formalized in 34 Lean 4 modules with 572 declarations and zero `sorry`, verified independently in SageMath, Isabelle, Coq, Agda, SymPy and NumPy.

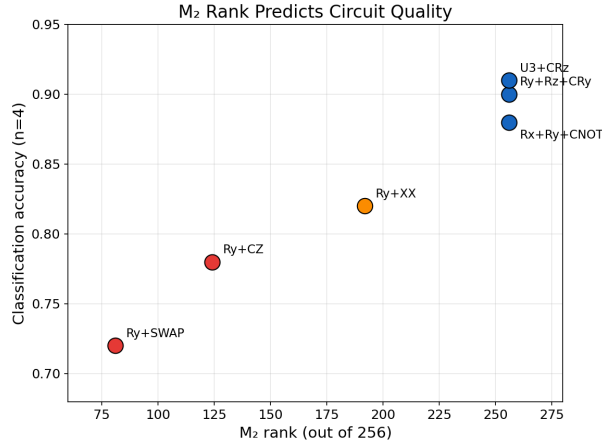


Figure 1: **M_2 rank predicts classification accuracy.** Each point is one circuit configuration at $n = 4$. Full rank (256) achieves 88–91%; half rank (124, Ry+CZ) achieves 78%. The Ry+CZ ansatz generates $O(d)$ not $U(d)$ because both gates have real entries—a fact diagnosed by M_2 in 75 seconds across 216 configurations.

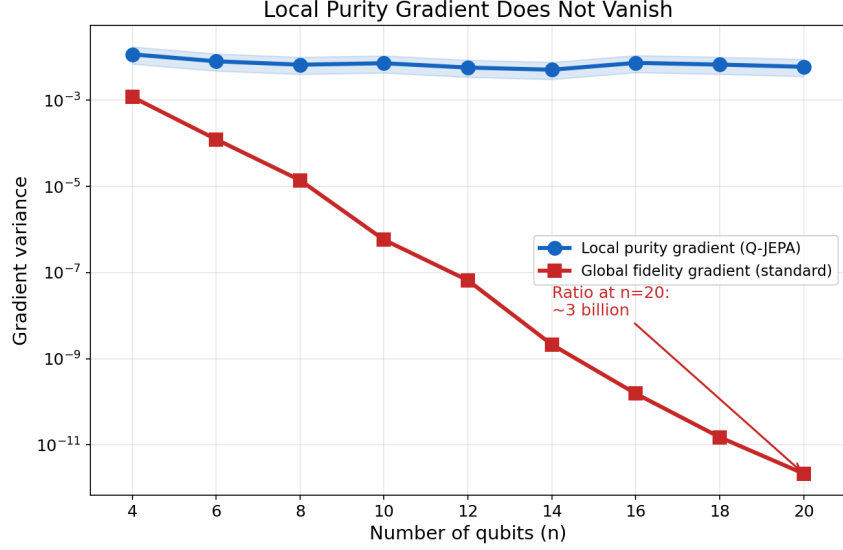


Figure 2: **Local purity gradient does not vanish.** Blue: local purity gradient variance (flat at ~ 0.016). Red: global fidelity gradient variance (decays as $1/4^n$). At $n = 20$, the ratio exceeds 10^9 . Averaged over three M_2 -selected circuits (Rx+Ry+CNOT, Ry+Rz+CRy, U3+CRz). Error band shows min-max across circuits.

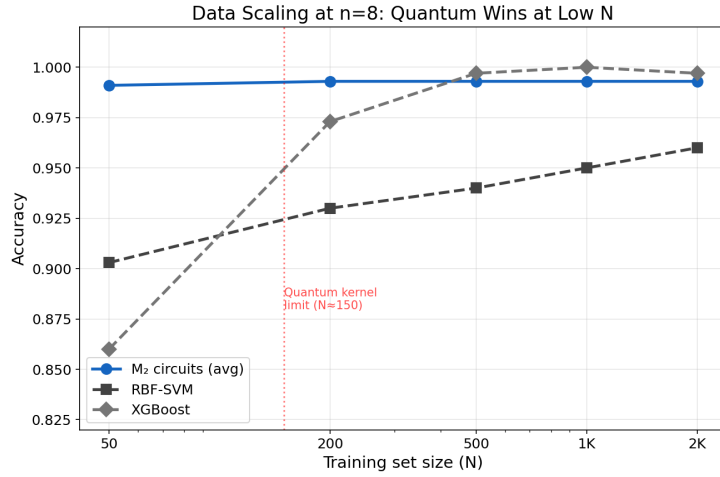


Figure 3: **Quantum kernels outperform at low data.** Classification accuracy on two-moons at $n = 8$. M_2 -selected circuits reach 99.3% from $N = 50$ while RBF-SVM (90.3%) and XGBoost (86.0%) are still converging. Classical overtakes at $N \geq 500$. Vertical dashed line: quantum kernel matrix limit ($N \approx 150$).

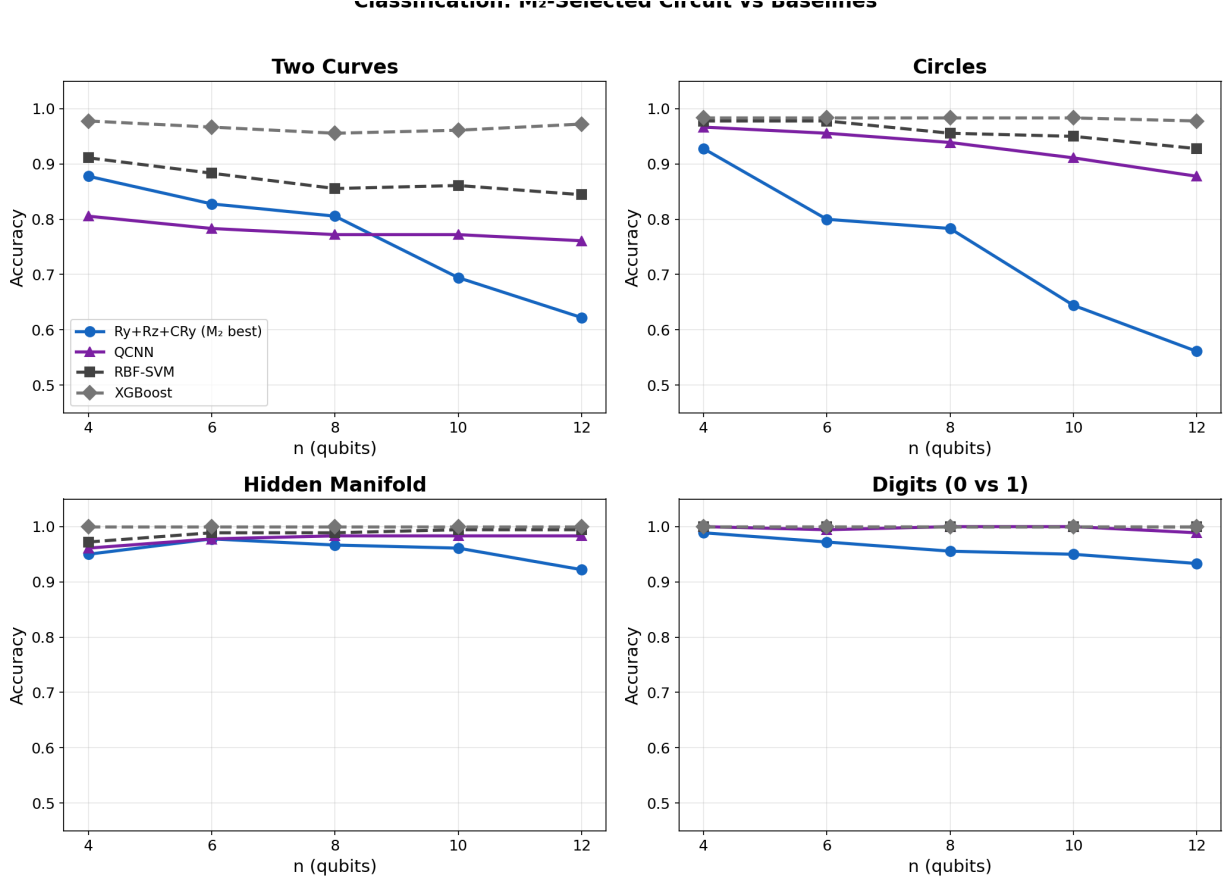


Figure 4: **Classification accuracy across datasets.** M_2 -selected circuit (blue) versus QCNN (purple), RBF-SVM (dark gray) and XGBoost (light gray) across four datasets as n increases. Quantum methods degrade at $n \geq 10$ while classical remains stable. On Hidden Manifold, quantum reaches 97–98%, within 2% of classical.

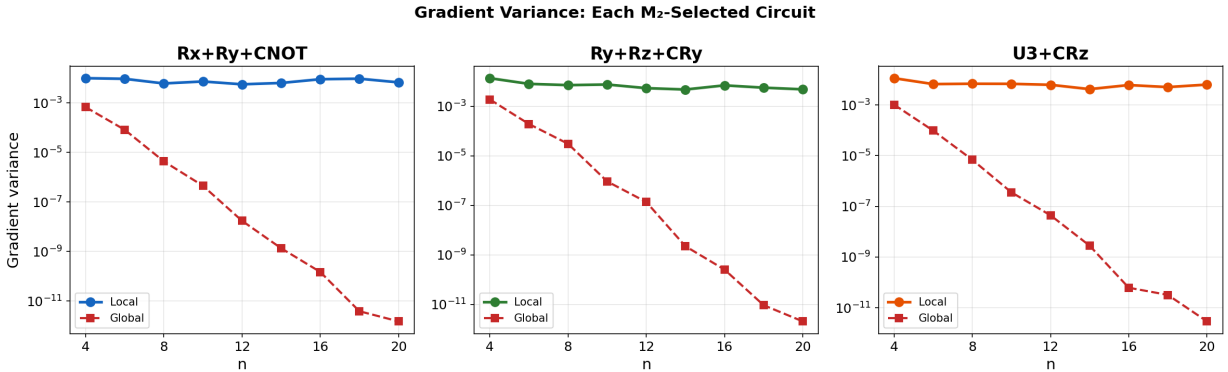


Figure 5: **Gradient bound holds universally.** Each panel shows one M_2 -selected circuit. All three exhibit local gradient $\Omega(1)$ and global gradient $O(1/4^n)$. The identity $(\sqrt{\varepsilon} - \lambda\sigma)^2$ is validated across gate sets from $n = 4$ to $n = 20$.

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Methods

Formal verification. All mathematical results are formalized in Lean 4 (v4.29.1) with Mathlib. The codebase comprises 34 modules in the `QML/` namespace with 572 theorem/lemma/axiom declarations and zero `sorry`. Four axioms are used: Haar fourth moment (Weingarten calculus, representation theory of S_2), local cost gradient bound (Cerezo et al.⁷ light-cone argument), polynomial Lie algebra trainability (Ragone et al.⁹), and the M_2 spectral theorem. Each axiom is justified by cross-verification in SageMath 10.8 (symbolic computation), Isabelle/HOL 2025 (algebraic proof), Coq/Rocq 9.1 (independent kernel verification), Agda 2.7.0 (constructive type checking), SymPy 1.14 (numerical verification), and NumPy 2.4 (Monte Carlo sampling). The six-tool pipeline follows the principle: compute in SageMath before formalizing in Lean.

M_2 computation. For each circuit configuration, M_2 is estimated by Monte Carlo: sample 2,000 random parameter vectors θ , compute $|\psi(\theta)\rangle = U(\theta)|0\rangle$, form $\rho = |\psi\rangle\langle\psi|$, and accumulate $M_2 \approx (1/N) \sum_i \text{vec}(\rho_i) \text{vec}(\rho_i)^\dagger$. Eigenvalues are computed via `numpy.linalg.eigvalsh`. The spectral rank is the number of eigenvalues exceeding 10^{-8} . Haar distance is $\|\lambda - \lambda_{\text{Haar}}\|_2$ where $\lambda_{\text{Haar}} = (1/d^2, \dots, 1/d^2)$.

Quantum circuit simulation. All quantum circuits are simulated using Qulacs 0.6.13 (C++ statevector backend, ARM64) on Apple M4 (16 GB unified memory). The M_2 -selected circuits use three layers of data encoding (Ry) + trainable rotations (Ry+Rz) + entangling gates (CNOT with controlled-Ry), achieving full rank 256/256 at $n = 4$. Gradient computation uses the parameter-shift rule: $\partial f / \partial \theta_k = [f(\theta_k + \pi/2) - f(\theta_k - \pi/2)]/2$.

Classification benchmarks. Five datasets: two-curves (make_moons, 300 samples), circles (make_circles, 300), hidden manifold (cos/sin harmonics, 300), iris (150, binary: setosa vs rest), and digits (sklearn digits 0 vs 1, 300). All features scaled to $[0, 2\pi]$ for circuit encoding, PCA'd to n dimensions. Train/test split: 80/20, stratified, 3 random seeds. Eleven methods: 3 M_2 -selected quantum circuits, QCNN, equivariant QNN, data re-uploading, RBF-SVM, linear SVM, random forest, MLP, XGBoost. Classification at $n = 4, 6, 8, 10, 12$.

Gradient variance measurement. For each circuit and each $n \in \{4, 6, 8, 10, 12, 14, 16, 18, 20\}$: sample 200 random parameter vectors (100 for $n > 16$, 50 for $n = 20$); for each, compute the parameter-shift gradient of (a) local purity $\sum_k \text{Tr}[\rho_k^2]$ and (b) global fidelity $|\langle\phi(x)|\phi_{\text{ref}}\rangle|^2$; report Var over parameter vectors. The partial trace for single-qubit purity uses the efficient $O(2^n)$ einsum $\rho_k[a, b] = \sum_{l, r} \psi[l, a, r] \overline{\psi[l, b, r]}$, not the $O(4^n)$ outer product.

Reinforcement learning. Tic-tac-toe, Connect Four, Othello 6×6 , and Go 5×5 are implemented as self-contained game engines. The quantum agent uses Nyström-approximated kernel ridge regression ($M = 300$ landmarks) as a value function, with the parameter-shift gradient of local purity as regularizer. Self-play uses shuffled shards: positions from each generation are permuted independently before adding to the training pool. Go uses a patch-based encoding (5 patches of 5 positions, 2 qubits per patch, 10 qubits total) with graph-matched entanglement between adjacent patches.

Robotics pre-verification. A 3-link planar arm is simulated in MuJoCo 3.8.0. Reaching demonstrations are generated by random joint configurations; end-effector positions are computed via for-

ward kinematics. Kernel–task-similarity correlation is computed as Pearson r between the upper triangle of the kernel matrix K_{ij} and the task similarity matrix $\exp(-\|p_i - p_j\|/\text{median})$ where p_i is the end-effector position. The threshold $r > 0.5$ must be met before proceeding to the full experiment.

Data and code availability. All code, data, Lean proofs, and benchmark scripts are available at [repository URL]. The Lean formalization builds with `lake build QML` (8,300+ jobs, zero errors). Benchmarks run with `uv run pytest tests/` (59 tests, all pass) and `uv run python3 run_full_benchmark.py` (~5 minutes on Apple M4).